Day 10 | Monday

10

November 16, 2026

Fraction addition and subtraction: job practice

12:45 PM–2:00 PM • Math • 75 min

Fraction cut lists, centre lines and kerf

2:15 PM–3:30 PM • Math • 75 min

Week 4 • 2 scheduled blocks • 150 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

12:45 PM–2:00 PM • 75 min

Fraction addition and subtraction: job practice

12:45 PM–12:50 PM	**Start**	5
12:50 PM–12:55 PM	**Recall**	5
12:55 PM–1:10 PM	**Teach**	15
1:10 PM–1:25 PM	**Together**	15
1:25 PM–1:45 PM	**Practise**	20
1:45 PM–1:55 PM	**Check**	10
1:55 PM–2:00 PM	**Wrap up**	5

Math • Math Week 4, Lesson 1

2:15 PM–3:30 PM • 75 min

Fraction cut lists, centre lines and kerf

2:15 PM–2:20 PM	**Start**	5
2:20 PM–2:25 PM	**Recall**	5
2:25 PM–2:40 PM	**Teach**	15
2:40 PM–2:55 PM	**Together**	15
2:55 PM–3:15 PM	**Practise**	20
3:15 PM–3:25 PM	**Check**	10
3:25 PM–3:30 PM	**Wrap up**	5

Math • Math Week 4, Lesson 2

Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time

Fraction addition and subtraction: job practice

Today’s goal: Add or subtract mixed-inch sizes using common denominators and record the result in lowest terms.

Share a prior use of this skill; identify the quantity and unit.

Textbook sections 4.3–4.6 pp.59–66

Exercise Book 4A Q10–15 (p.26); 4B Q2–3 (p.27)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 4.3–4.6 pp.59–66

Recall and reconnect

Use the Week 3 checks to review common denominators and regrouping. Start with Exercise Book 4A questions 6 and 10.

Attempt the short retrieval task independently; explain the method before seeing answers.

Resources: Exercise Book 4A Q10–15 (p.26); 4B Q2–3 (p.27)

Textbook sections 4.3–4.6 pp.59–66

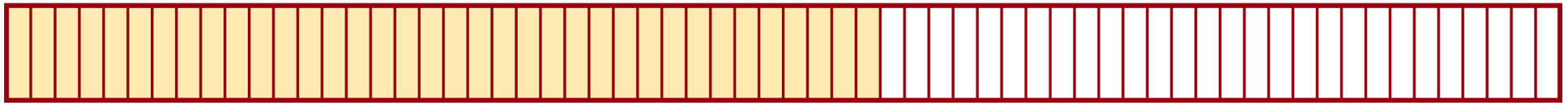
Recall from the Exercise Book | 4A Q6

Write $\frac{14}{32}$ and $\frac{36}{64}$ in lowest terms.

14/32



36/64

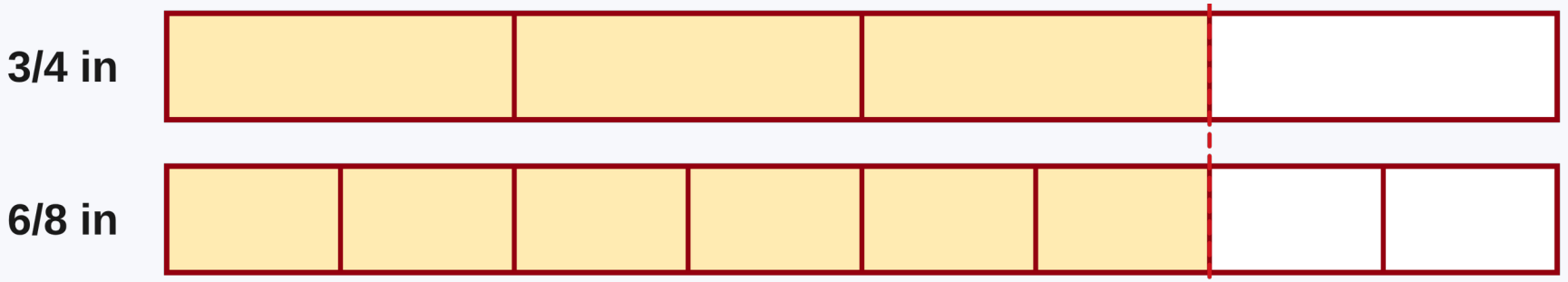


Write the operation, working and units before checking.

Textbook sections 4.3–4.6 pp.59–66

Add lengths with matching parts

1. Use one unit: write every length in inches before combining it.
2. Choose a common denominator so both fractions name equal-sized parts.
3. Rewrite equivalent fractions; add numerators and keep the common denominator.
4. Combine whole inches, carry any whole from the fraction, then reduce.



Textbook sections 4.3–4.6 pp.59–66

Subtract across a whole inch

1. Compare the fractional parts before you subtract.
2. If needed, trade one whole inch for a full denominator of fractional parts.
3. Subtract whole and fractional parts; simplify the result.
4. Check by adding the result to the length you took away.



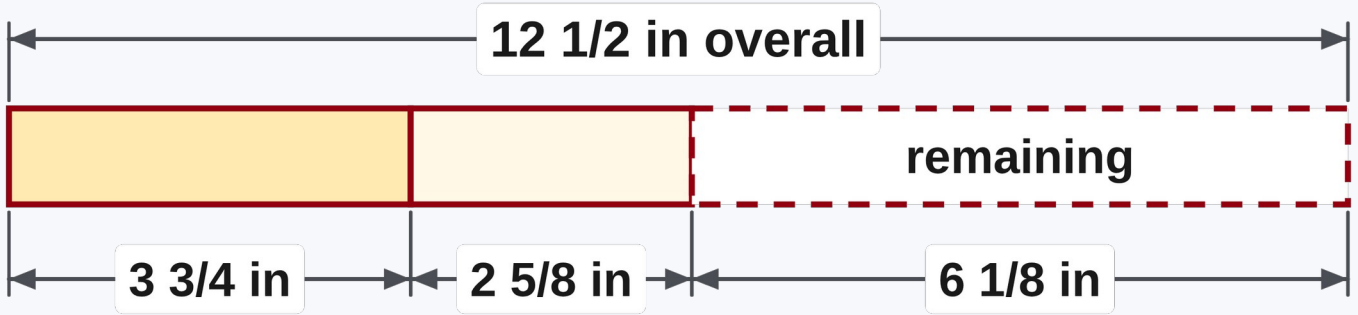
Textbook sections 4.3–4.6 pp.59–66

Use the overall dimension as a check

Classroom sketch: total 12 1/2 in; known parts 3 3/4 in and 2 5/8 in.

Known = 3 6/8 + 2 5/8 = 6 3/8 in
Remaining = 12 4/8 − 6 3/8 = 6 1/8 in

Check: 6 3/8 + 6 1/8 = 12 1/2 in.
Mark the fractional part on a rule.



Together: 4A Q14(a), Q15(a), then 4B Q2. Explain regrouping to a partner; check fraction entry on the selected calculator.

Textbook | page 65

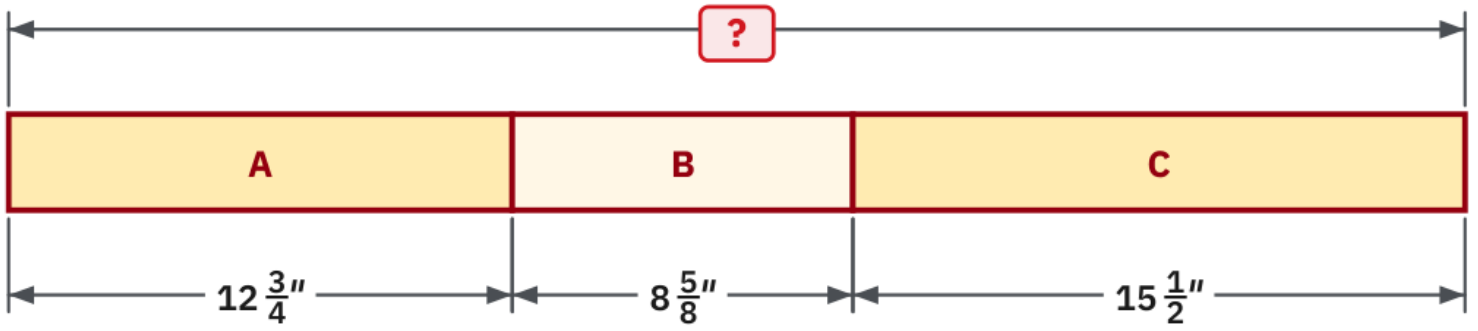


Figure 4.9 Three pieces on a cut list. The total length is hidden.

Textbook p.65 • Figure 4.9 — Fractional cut-list dimensions

Textbook sections 4.3–4.6 pp.59–66

Textbook | page 65

Example 4.9 Total of a cut list

Find the total length of the three pieces: $12\frac{3}{4}$ in, $8\frac{5}{8}$ in, $15\frac{1}{2}$ in.

SOLUTION

1. Add the whole numbers: 12 + 8 + 15 = 35.
2. Change the fractions to eighths and add them: $\frac{6}{8} + \frac{5}{8} + \frac{4}{8} = \frac{15}{8}$.
3. $\frac{15}{8} = 1\frac{7}{8}$. Carry the 1: 35 + 1 = 36. The total is $36\frac{7}{8}$ in.

Textbook p.65 • Example 4.9 — Common denominator and carrying a whole inch

Textbook sections 4.3–4.6 pp.59–66

Spot the mistake. Explain the correction.

A learner adds the denominators when adding two fractional lengths. What must stay the same?

Discuss the method before checking the answer.

The denominator names the size of each part. Rename with a common denominator; add only numerators.

Textbook sections 4.3–4.6 pp.59–66

Work together

Work Exercise Book 4A Q14(a) and Q15(a), then Exercise Book 4B Q2. Explain the regrouping to a partner.

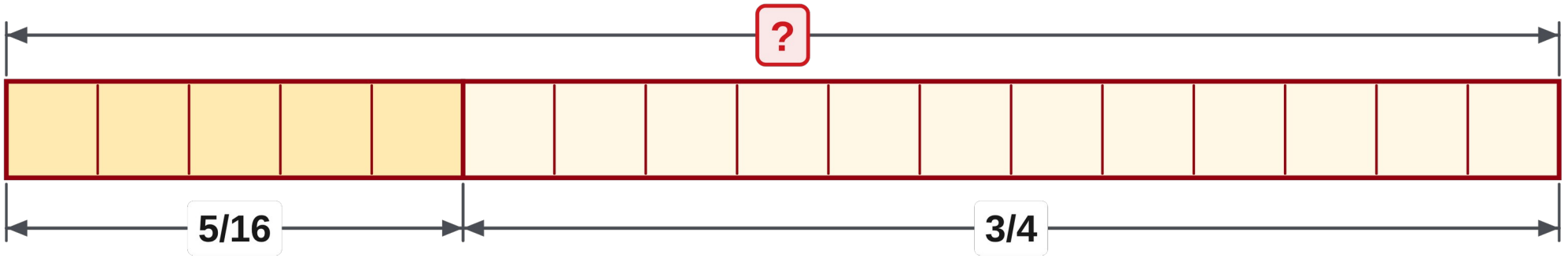
Work in pairs; alternate calculator operator and checker. Show units and an estimated answer.

Resources: Exercise Book 4A Q10–15 (p.26); 4B Q2–3 (p.27)

Textbook sections 4.3–4.6 pp.59–66

Guided practice | 4A Q14a

Add. (a) $\frac{5}{16} + \frac{3}{4}$



Work with a partner. Take turns explaining the next step.

Textbook sections 4.3–4.6 pp.59–66

Compare your method | 4A Q14a

Add. (a) $\frac{5}{16} + \frac{3}{4}$

CHECK TOGETHER

(a) 1 1/16

Textbook sections 4.3–4.6 pp.59–66

Your scheduled practice

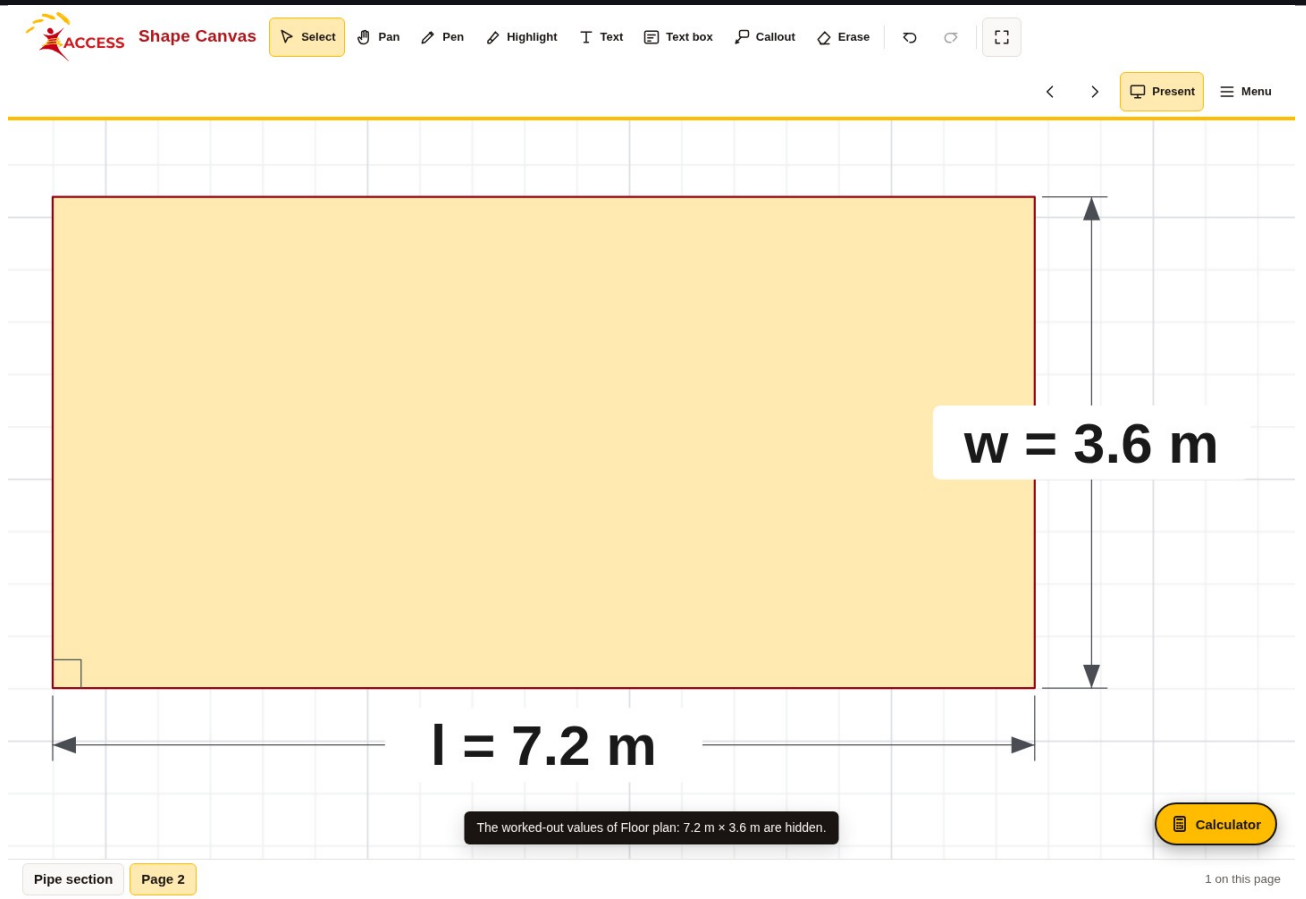
Complete remaining Exercise Book 4A Q14–15 and 4B Q3. Mark each result on a paper rule; use calculator checking only after the written solution.

Solve on paper first; use the tool to check; correct one error with a written reason.

Resources: Exercise Book 4A Q10–15 (p.26); 4B Q2–3 (p.27)

Textbook sections 4.3–4.6 pp.59–66

Open Shape Canvas



Navigation example. Use the values and units in the assigned task.

Textbook sections 4.3–4.6 pp.59–66

Practise in Numeracy Hub and Shape Canvas

1. Set Shape Canvas to in and enter the fractional rectangle dimensions from today’s work.

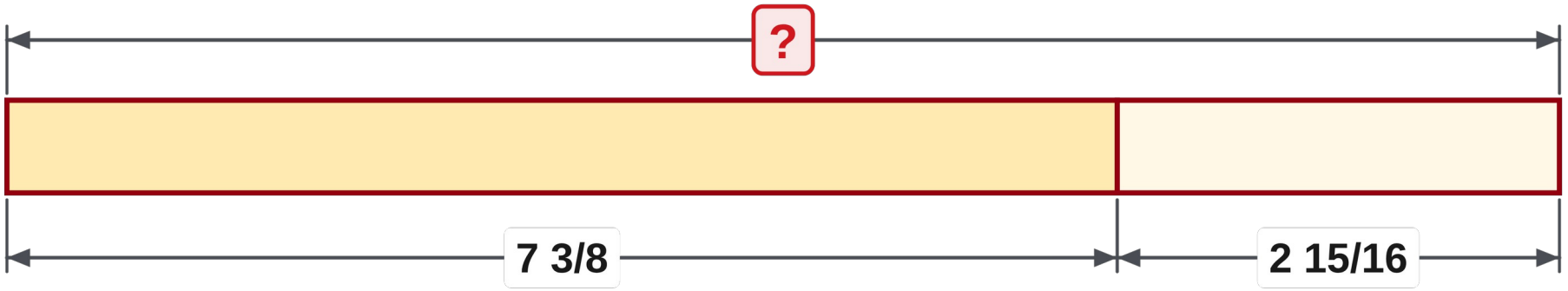
2. Calculate the assigned addition or subtraction on paper; mark the result on a paper rule.

3. Use the selected Numeracy Hub calculator to check the fraction entry and compare results.

Textbook sections 4.3–4.6 pp.59–66

Practise from the Exercise Book | 4A Q14b

Add. (b) $7\frac{3}{8} + 2\frac{15}{16}$

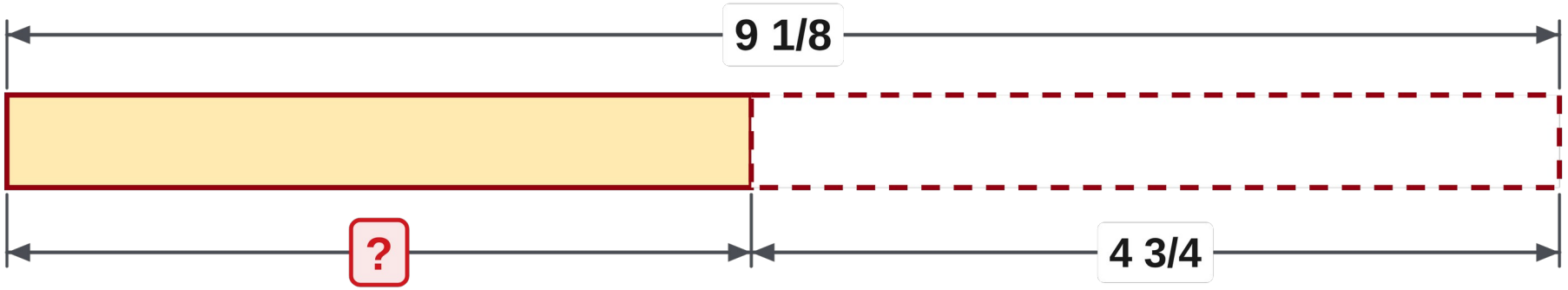


Write the operation, working and units before checking.

Textbook sections 4.3–4.6 pp.59–66

Practise from the Exercise Book | 4A Q15b

Subtract. (b) $9\frac{1}{8} - 4\frac{3}{4}$



Write the operation, working and units before checking.

Textbook sections 4.3–4.6 pp.59–66

Check independently

Exercise Book 4A Q14(b) and Q15(b). Show common denominators and simplified answers.

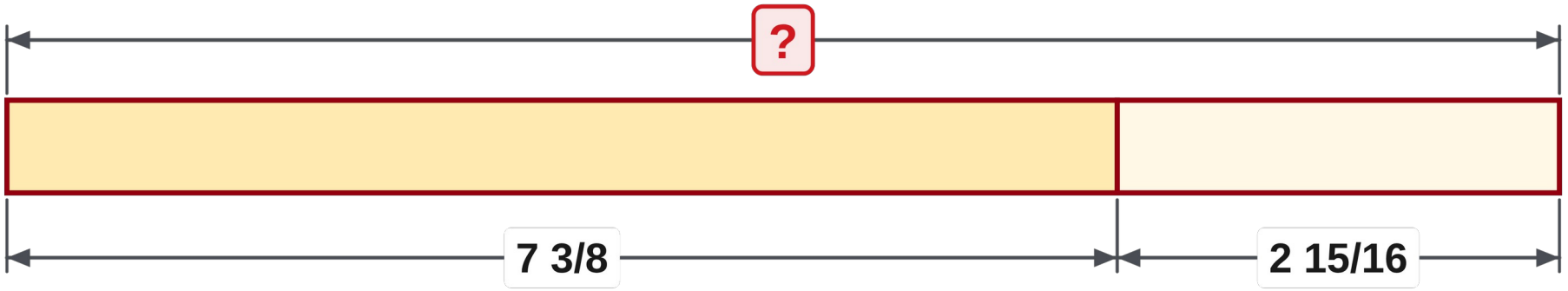
Solve without displayed answers; include working, units and the requested rounding.

Resources: Exercise Book 4A Q10–15 (p.26); 4B Q2–3 (p.27)

Textbook sections 4.3–4.6 pp.59–66

Independent check | 4A Q14b

Add. (b) $7\frac{3}{8} + 2\frac{15}{16}$

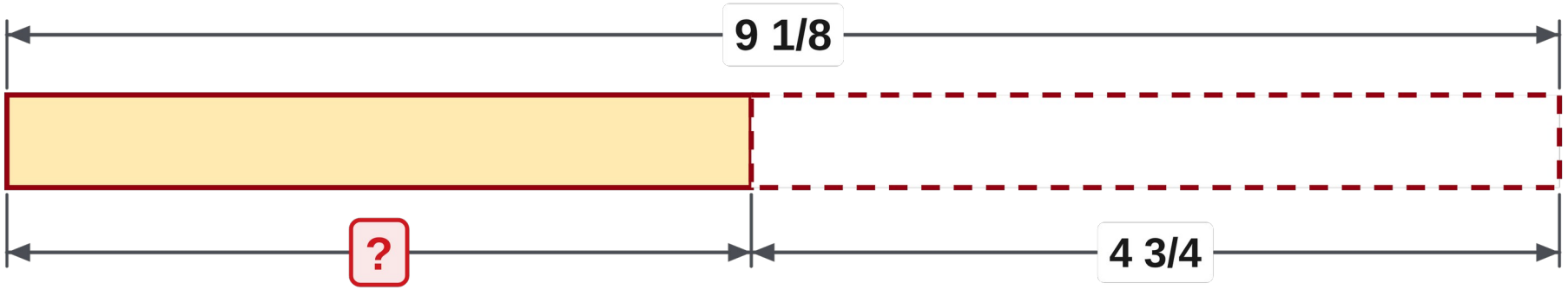


Answer independently. Keep worked solutions closed.

Textbook sections 4.3–4.6 pp.59–66

Independent check | 4A Q15b

Subtract. (b) $9\frac{1}{8} - 4\frac{3}{4}$



Answer independently. Keep worked solutions closed.

Textbook sections 4.3–4.6 pp.59–66

Wrap up and choose the next step

Ask: What did you learn? Where will you use it? What will you check next time?

Name the math and one other essential skill used; write the next practice step.

Resources: Exercise Book 4A Q10–15 (p.26); 4B Q2–3 (p.27)

Textbook sections 4.3–4.6 pp.59–66

Break

2:00 PM–2:15 PM

Return ready for: Fraction cut lists, centre lines and kerf

This break is excluded from teaching minutes.

Fraction cut lists, centre lines and kerf

Today’s goal: Use multiplication and division of fractions to find repeated cuts, centres and offcuts including stated kerf.

Share a prior use of this skill; identify the quantity and unit.

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Exercise Book 4A Q16–17 (p.26); 4B Q1,4–7 (pp.27–28); 4C Q5 (p.30)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Recall and reconnect

Explain how many equal gaps lie between nine marked points. Review the difference between a piece and a saw cut.

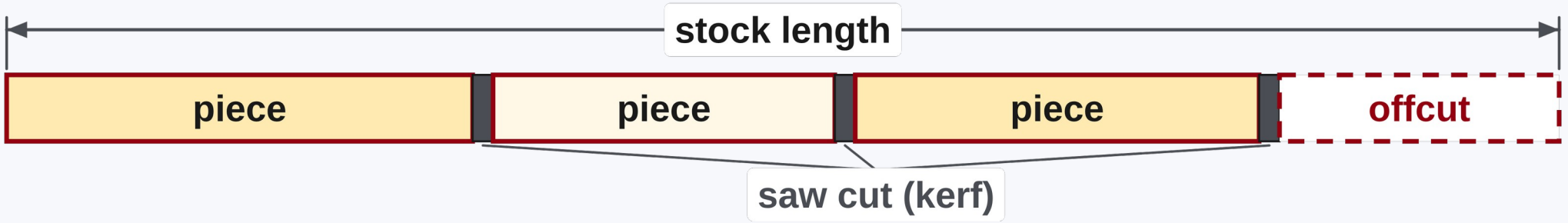
Attempt the short retrieval task independently; explain the method before seeing answers.

Resources: Exercise Book 4A Q16–17 (p.26); 4B Q1,4–7 (pp.27–28); 4C Q5 (p.30)

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Count pieces, spaces and cuts

1. Draw the stock and label every required piece before calculating.
2. Total repeated pieces = number of pieces × length of one piece.
3. Count the saw cuts shown in the problem; include each stated kerf.
4. Offcut = stock length − piece lengths − total stated kerf.



Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

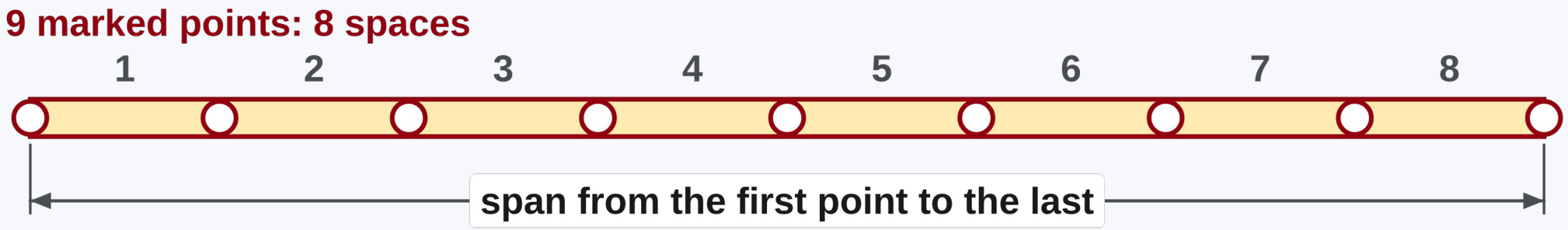
Divide lengths into equal parts

1 Nine marked points have eight spaces between the first and last point.

2 For equal spacing, divide the span by the number of spaces.

3 Divide by a fraction by multiplying by its reciprocal.

4 Round down for complete pieces; keep the remainder as an offcut.



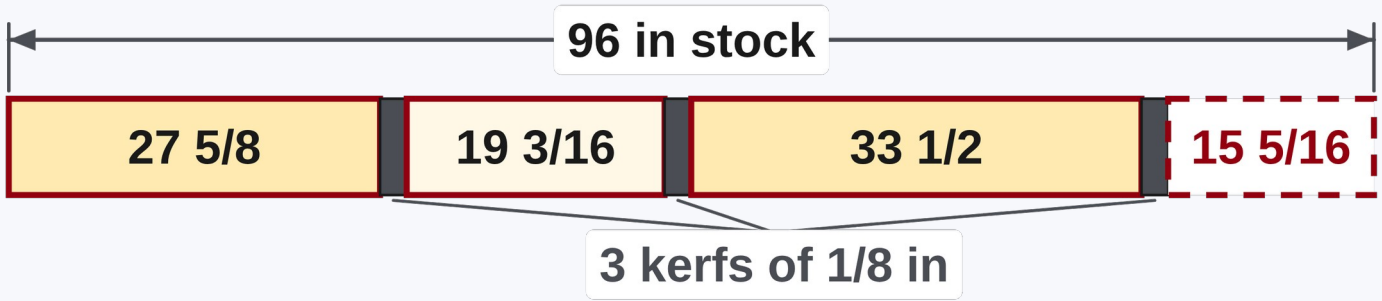
Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Keep cutting loss inside the used total

Exercise Book 4B Q1: 96 in stock; three stated 1/8 in kerfs.

Pieces = 27 5/8 + 19 3/16 + 33 1/2 = 80 5/16 in
Kerf = 3 × 1/8 = 3/8 = 6/16 in
Offcut = 96 − 80 11/16 = 15 5/16 in

Check: pieces + kerf + offcut = 96 in.



Together: 4B Q4–5. Draw equal spaces for Q6 before writing an equation. Shape Canvas: repeated rectangles plus stock and kerf labels.

W4 L2 • Given kerf is problem data, not a universal allowance

Textbook | page 67

SAW CUTS IN FRACTIONS

When a problem gives the **kerf** as a fraction of an inch, work with it like any other fraction. With a kerf of $\frac{1}{8}$ in, 4 cuts remove $4 \times \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$ in of stock. The problems in this book give the kerf whenever it counts.

Textbook p.67 • Saw cuts in fractions — Count the stated kerfs

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Textbook | page 67

Example 4.11 Pieces and saw cuts

Four pieces, each $11\frac{3}{4}$ in long, are cut from a 48 in bar. You make 4 cuts and the kerf is $\frac{1}{8}$ in. How long is the offcut?

SOLUTION

1. Pieces: $4 \times 11\frac{3}{4} = 4 \times \frac{47}{4} = \frac{188}{4} = 47$ in.
2. Saw cuts: $4 \times \frac{1}{8} = \frac{1}{2}$ in.
3. Offcut: $48 - 47 - \frac{1}{2} = \frac{1}{2}$ in.

Textbook p.67 • Example 4.11 — Pieces, kerf and offcut

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Spot the mistake. Explain the correction.

A cut list has three pieces. Can you assume there are always three cuts?

Discuss the method before checking the answer.

No. Draw and count the cuts required by that question. Use only its stated kerf.

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Work together

Solve Exercise Book 4B Q4–5 together. Draw the equal spaces for Q6 before forming the equation.

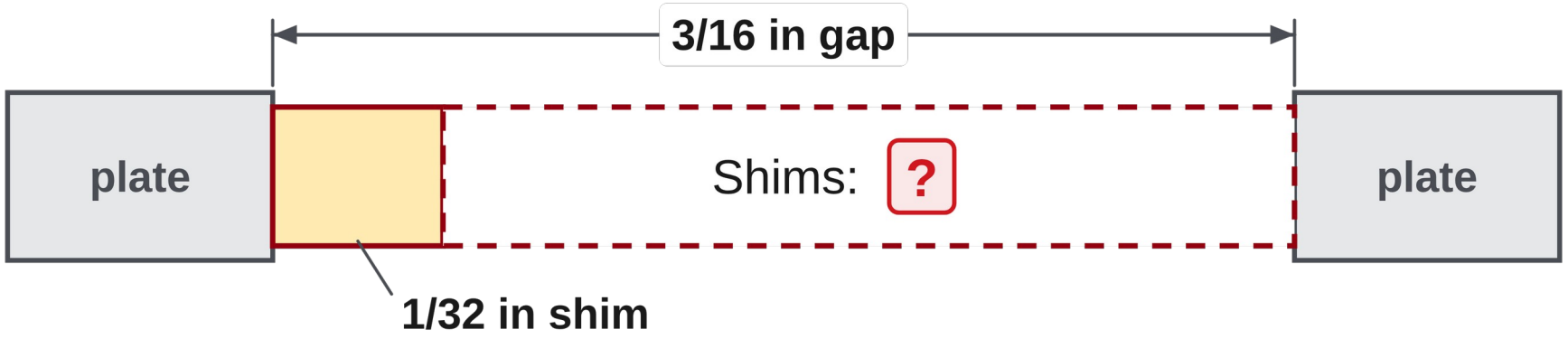
Work in pairs; alternate calculator operator and checker. Show units and an estimated answer.

Resources: Exercise Book 4A Q16–17 (p.26); 4B Q1,4–7 (pp.27–28); 4C Q5 (p.30)

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Guided practice | 4B Q4

A millwright needs to fill a $\frac{3}{16}$ in gap with $\frac{1}{32}$ in shims. How many shims does it take?



Work with a partner. Take turns explaining the next step.

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Compare your method | 4B Q4

A millwright needs to fill a $\frac{3}{16}$ in gap with $\frac{1}{32}$ in shims. How many shims does it take?

CHECK TOGETHER

6 shims

3/16 ÷ 1/32 = 3/16 × 32/1 = 6

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Your scheduled practice

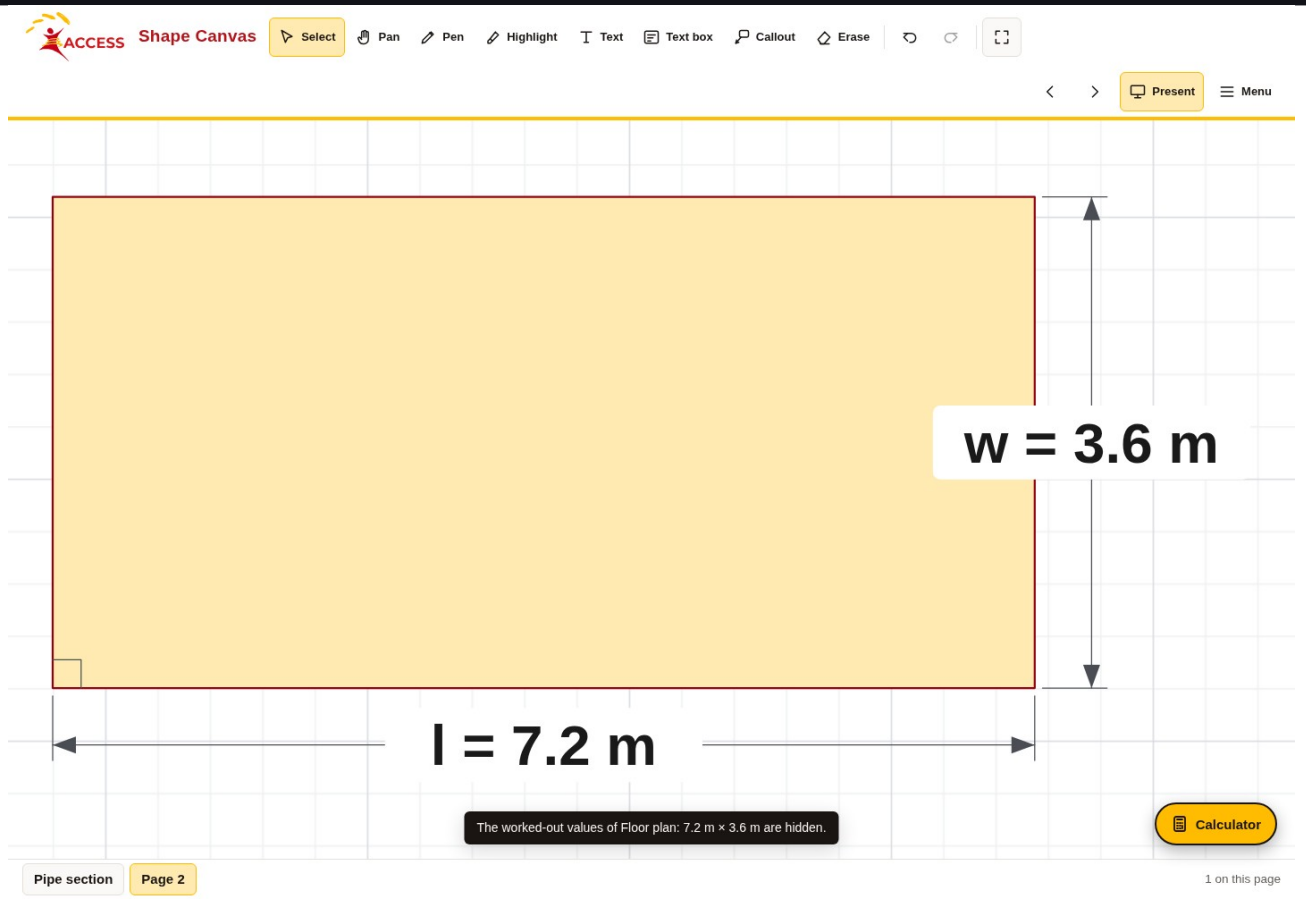
Solve Exercise Book 4A Q16–17, 4B Q6–7 and 4C Q5 as time permits. Explain why complete pieces round down; use only the kerf given in the question.

Solve on paper first; use the tool to check; correct one error with a written reason.

Resources: Exercise Book 4A Q16–17 (p.26); 4B Q1,4–7 (pp.27–28); 4C Q5 (p.30)

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Open Shape Canvas



Navigation example. Use the values and units in the assigned task.

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Practise in Numeracy Hub and Shape Canvas

1. In Shape Canvas, use in units and repeat rectangles for the assigned cut-list pieces.

2. Add text labels for stock length and the problem’s stated kerf; solve on paper.

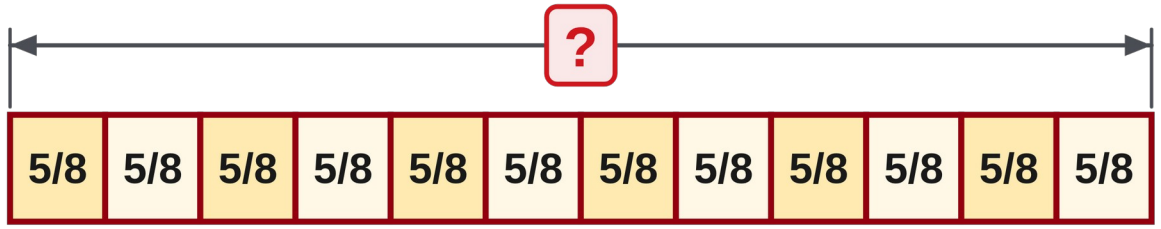
3. Check the total and offcut with Numeracy Hub Calculator after the written solution.

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

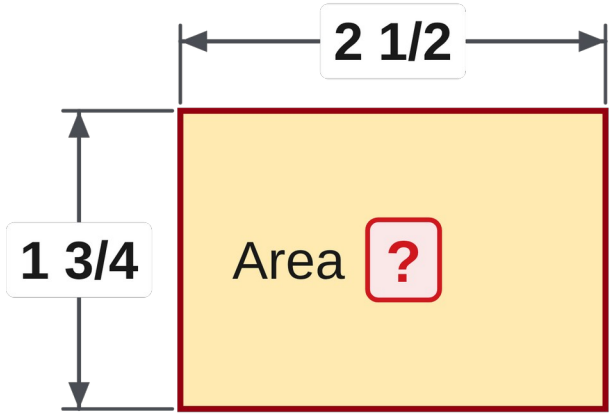
Practise from the Exercise Book | 4A Q16

Multiply. (a) $\frac{5}{8} \times 12$ (b) $2\frac{1}{2} \times 1\frac{3}{4}$

(a)



(b)



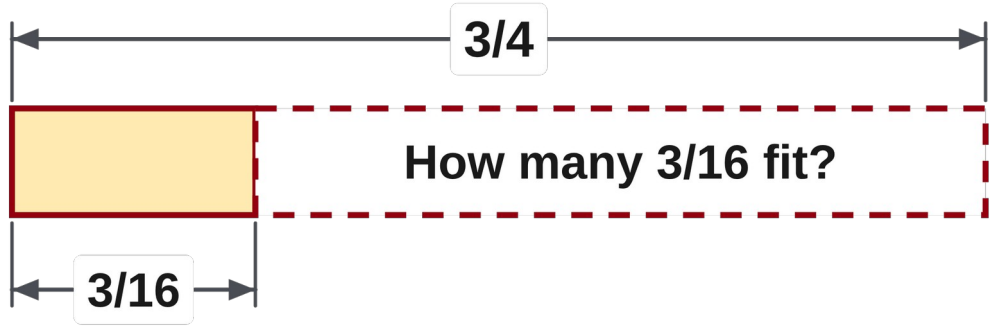
Write the operation, working and units before checking.

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

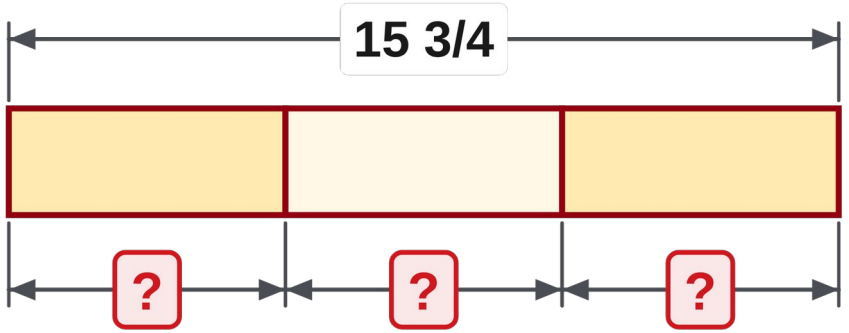
Practise from the Exercise Book | 4A Q17

Divide. (a) $\frac{3}{4} \div \frac{3}{16}$ (b) $15\frac{3}{4} \div 3$

(a)



(b)



Write the operation, working and units before checking.

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Check independently

Exercise Book 4B Q5 and Q7; show centre line, complete pieces and offcut.

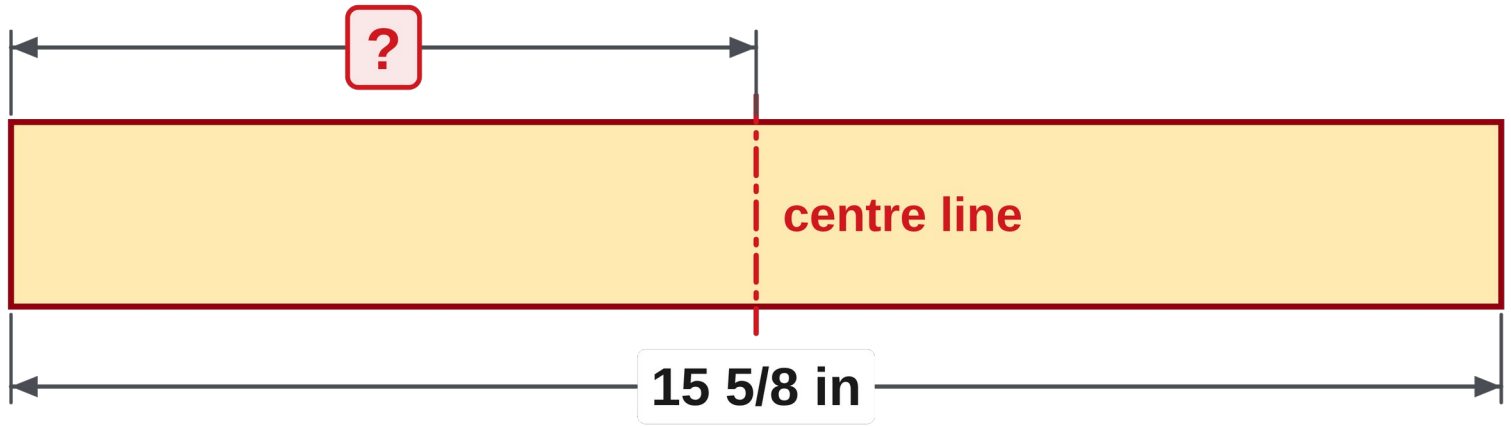
Solve without displayed answers; include working, units and the requested rounding.

Resources: Exercise Book 4A Q16–17 (p.26); 4B Q1,4–7 (pp.27–28); 4C Q5 (p.30)

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Independent check | 4B Q5

A plate is $15\frac{5}{8}$ in wide. How far from the edge is the centre line?

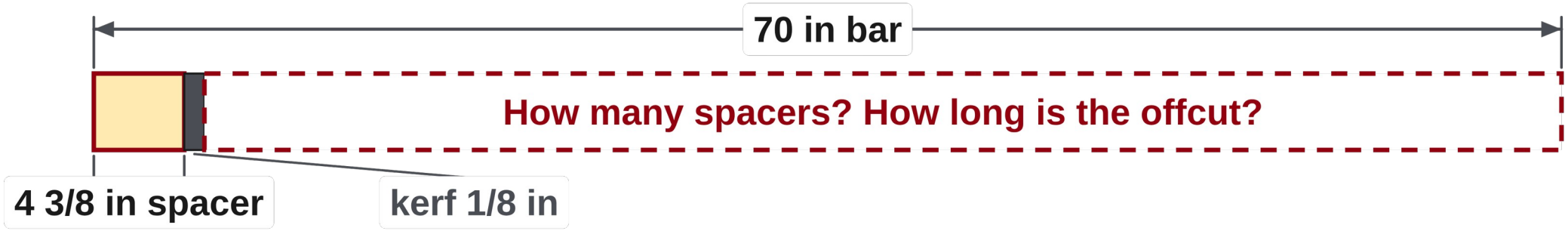


Answer independently. Keep worked solutions closed.

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Independent check | 4B Q7

How many $4\frac{3}{8}$ in spacers can be cut from a 70 in bar? You make one cut for each spacer and the kerf is $\frac{1}{8}$ in. How long is the offcut?



Answer independently. Keep worked solutions closed.

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74

Wrap up and choose the next step

Ask: What did you learn? Where will you use it? What will you check next time?

Name the math and one other essential skill used; write the next practice step.

Resources: Exercise Book 4A Q16–17 (p.26); 4B Q1,4–7 (pp.27–28); 4C Q5 (p.30)

Textbook sections 4.7 pp.67–70; fraction worksheets pp.71–74